

Interest Rate Models for Quants

PnL, Hedging, and Model Choice Across the Yield Curve
Why Desks Choose Hull–White, LMM, SABR — and When They Break

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Example: Long \$100M 10Y payer swaption - Gamma = \$50,000 per bp² - Rates move 20bp → $PnL_{\Gamma} = \frac{1}{2} \times 50000 \times 20^2 = \$10M$

Sign interpretation: - Long gamma: Win on large moves (any direction) - Short gamma: Lose on large moves → ****dangerous in crisis****

Vanna: Delta-Vol Cross-Effect

Definition:

$$\text{Vanna} = \frac{\partial^2 V}{\partial r \partial \sigma} = \frac{\partial \Delta}{\partial \sigma}$$

What it means: - If vol rises, your delta changes - ****Classic trap:**** Hedge delta today, vol spikes, delta hedge is now wrong

Example: - Delta-hedged swaption book - Vol spikes 10 points → Delta shifts by vanna × 10 - Unhedged delta creates \$2M loss

How to hedge: Buy/sell options to offset vanna (e.g., risk reversals)

Volga: Vol Convexity

Definition:

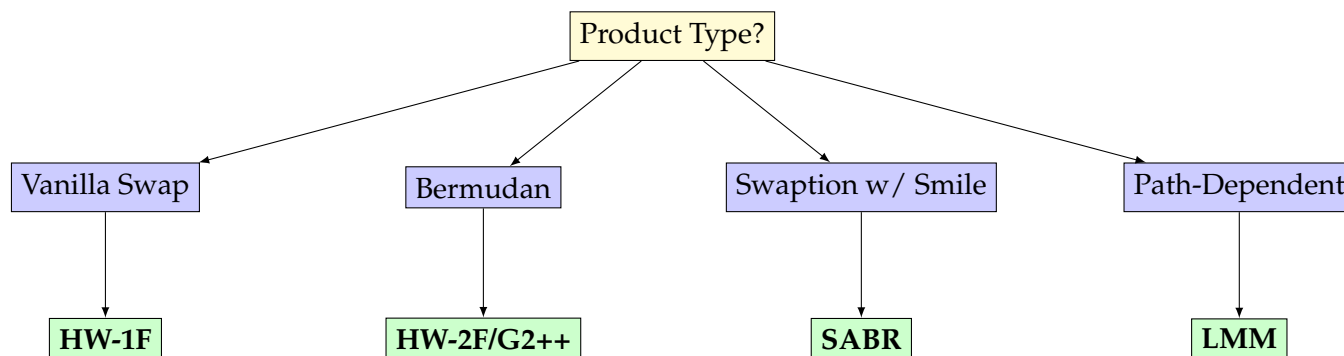
$$\text{Volga} = \frac{\partial^2 V}{\partial \sigma^2} = \frac{\partial \text{Vega}}{\partial \sigma}$$

What it means: - Vega changes as vol changes - Long volga: Vega increases when vol rises → ****accelerating gains****

Interview: "After delta-hedging, I check gamma, vanna, volga. If volga is large and negative, I'm short vol convexity—dangerous if vol spikes further."

Interview Decision Trees

Tree 1: "What Model Would You Use?"



Interview Script: "First I identify the product type, then the dominant risk. Vanilla → DV01 → HW. Bermudan → exercise boundary → G2++. Smile-sensitive → SABR. Path-dependent → LMM."

Top 10 Interview Mistakes

Don't Make These Mistakes

1. **Saying "CIR for negative rates"** → CIR BREAKS at negative rates
2. **Using "risk-neutral" without context** → Specify which measure
3. **Ignoring calibration** → Always mention what you calibrate to
4. **Overcomplexity** → "For this vanilla swap, I'd use LMM-SABR" ← NO!
5. **No economic intuition** → Always start with "why" before "how"
6. **Forgetting post-crisis changes** → LIBOR → SOFR, basis risk
7. **Saying "no arbitrage" without explaining** → Show the constraint
8. **Confusing forward and spot measures** → Be precise
9. **Not mentioning speed** → Desks care about computation time
10. **Missing the trade-off** → Every model has pros/cons

Mock Interview: 20-Minute Rates Grilling

Time yourself. Give answers out loud. This simulates real pressure.

Round 1: Warm-Up (5 min)

Q1: "Walk me through how you'd price a 10Y vanilla swap."

[Pause 60 seconds. Answer out loud.]

Model Answer: "I'd use HW-1F calibrated to today's yield curve. Extract discount factors from the curve, compute fixed leg PV as sum of discounted cashflows, floating leg is par. Swap value = Fixed PV - Floating PV. I hedge with DV01 computed from $B(t, T)$ sensitivity."

Follow-up: "Why HW and not LMM?" "Speed. HW has closed-form bond prices and trees. LMM needs Monte Carlo, overkill for vanilla DV01 risk."

Round 2: Technical (7 min)

Q2: "The yield curve inverts. What happens to your Bermudan swaption?"

[Pause 90 seconds.]

Model Answer: "Inversion means short rates expected to fall (recession pricing). For a payer Bermudan (right to enter pay-fixed), early exercise becomes less attractive—you'd rather wait for rates to fall further. So time value increases, optimal exercise moves later. In my G2++ model, I'd see the exercise boundary shift toward later dates."

Pressure Follow-up: "But doesn't inversion mean higher forward rates?" "No—inverted curve means near-term spot rates > long-term. Forward rates reflect expected future spots, which are lower. That's why expectations hypothesis fails."

Round 3: Calibration Risk (5 min)

Q3: "Your SABR calibration shows ν jumping from 0.3 to 0.8 overnight, but the swaption smile shape looks similar. Is this real market behavior or a calibration problem?"

[Pause 60 seconds.]

Model Answer: "This is almost certainly a **calibration problem**, not real market behavior. Vol-of-vol ν controls the curvature of the smile wings. A jump from 0.3 to 0.8 would imply a massive increase in tail risk, which would be visible as a much steeper smile. Since the smile looks similar, the optimizer is likely compensating for other miscalibrated parameters—maybe ρ or α —or the correlation matrix is near-singular, causing instability. I'd first check: (1) Are all swaption quotes liquid and reasonable? (2) Is the correlation matrix still PSD? (3) Did I fix β or let it float? The fix is to add a penalty term to the objective function or use the prior day's parameters as a stronger prior."

Pressure Follow-up: "But what if the 10Y point really did gap due to a central bank announcement?" "Then I'd see the **entire smile shift**, not just ν spike. I'd see α move (ATM vol level) and the forward rate move. A jump in ν alone is a calibration artifact. Real market events affect multiple parameters simultaneously, not just the wing curvature."

Round 4: Crisis Breakdown (4 min)

Q4: "You're short EUR swaptions when ECB cuts rates to -0.5[Pause 45 seconds.]

Model Answer: "The lognormal LMM assumes $F_i(t) > 0$ —it can't handle negative forward rates. When the ECB cut to -0.5

Pressure Follow-up: "Can't you just keep the lognormal model and shift rates up by 2?" "That's a **shifted lognormal** model, which is a valid quick fix but it's **ad-hoc**. The shift parameter β becomes a new source of model risk—if you set it to 2

Round 5: Exotic Product Design (6 min)

Q5: "A hedge fund wants a 30Y CMS spread option (10Y-2Y) that pays 10x the spread widenings. What model do you use, what's the dominant risk, and what reserve would you hold?"

[Pause 90 seconds.]

Model Answer: "I'd use **LMM with 2 factors minimum**—probably LMM-SABR to capture smile. The dominant risk is **forward-forward correlation breakdown** between the 2Y and 10Y rates, especially in crisis when both legs move together and the spread collapses. The 10x leverage magnifies tail risk. **Reserves:** (1) **Correlation risk reserve:** 15-20

Pressure Follow-up: "Why not just use HW-2F? It's faster and calibrates to swaptions." "HW-2F correlates all forwards perfectly within its two factors—it can't capture the **specific** 2Y-10Y correlation that decouples in stress. CMS spread option lives or dies on that correlation, so you need LMM's granular forward structure. HW-2F would underprice the tail risk where both legs correlate to 1 and the spread $\rightarrow 0$. Speed is useless if the model misses the primary risk."